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MECHANICAL ARM SYSTEM FOR COLLECTING GARBAGE FROM A GARBAGE CONTAINER

Abstract

The mechanical arm system can include a garbage container holder and at least one mast. The holder can have guiding wheels engaged with adjacent tracks of the mast, the two adjacent tracks both having a lower portion extending upwardly along the mast, the lower portion of the first track being located forwardly of the lower portion of the second track, the lower portion of both tracks leading into corresponding upper portions which are curved rearwardly. The system can have a primary mast and a secondary mast slidably mounted to the primary mast, with the garbage container slidably mounted to the secondary mast. A driving link can simultaneously connect the primary mast to the secondary mast, and the secondary mast to the holder in a manner that when the driving link is moved, both sliding movements are performed in the same direction.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application is a continuation of U.S. patent application Ser. No. 18/413,621, filed on Jan. 16, 2024, which is a continuation of U.S. patent application Ser. No. 17/736,461, filed on May 4, 2022, now U.S. Pat. No. 11,932,486, which is a continuation of U.S. patent application Ser. No. 16/635,757, filed on Jan. 31, 2020, now U.S. Pat. No. 11,332,308, which is a National Stage Application under 35 USC § 371 and claims the benefit of International Patent Application No. PCT/CA2018/050969, filed on Aug. 10, 2018, which claims the benefit of U.S. Provisional Patent Application No. 62/544,981, filed on Aug. 14, 2017, which are incorporated herein by reference in their entirety.

BACKGROUND

[0002] Known mechanical arms for performing automated collection of garbage bins and dumping of the content of the garbage bins in a container of a corresponding garbage collection truck have been satisfactory to a certain degree, but there remains room for improvement. For instance, some mechanical arms performing automated collection of garbage tend to suffer from efficiency or durability issues, i.e. they tend to be subject to high failure rates as a result of designs which were not strong enough to withstand the high number of cycles and the heavy loads involved and required more maintenance than what was desired. In some cases, mechanical arms which are designed to achieve high performances in terms of cycle time, versatility and/or durability, may suffer from high manufacturing costs.

[0003] In view of the above, there is a need for an improved mechanical arm for a garbage collection truck performing automated garbage collection from a garbage bin which, by virtue of its design and components, would be able to overcome or at least minimize some of the above-discussed prior art concerns.

SUMMARY

[0004] The mechanical arm system can include a garbage container holder and at least one mast. The holder can have guiding wheels engaged with adjacent tracks of the mast, the two adjacent tracks both having a lower portion extending upwardly along the mast, the lower portion of the first track being located forwardly of the lower portion of the second track, the lower portion of both tracks leading into corresponding upper portions which are curved rearwardly. The system can have a primary mast and a secondary mast slidably mounted to the primary mast, with the garbage container slidable mounted to the secondary mast. A driving link can simultaneously connect the primary mast to the secondary mast, and the secondary mast to the holder in a manner that when the driving link is moved, both sliding movements are performed in the same direction.

[0005] In accordance with one aspect, there is provided a mechanical arm system for collecting garbage from a garbage container, the mechanical arm system comprising: a garbage container holder having two guiding wheels at a rear thereof, the garbage container being held at a front of the garbage container holder; a mast having two adjacent tracks each receiving a corresponding wheel to guide the movement thereof, the two adjacent tracks both having a lower portion

extending upwardly along the mast, the lower portion of the first track being located forwardly of the lower portion of the second track, the lower portion of both tracks leading into corresponding upper portions which are curved rearwardly in a manner that the garbage container holder can be raised along the lower portion of the tracks and into the upper portion of the tracks, the garbage container being tilted upside down to empty its contents behind the mast as it is conveyed along the upper portions of the tracks.

[0006] In accordance with another aspect, there is provided a mechanical arm system for collecting garbage from a garbage container, the mechanical arm system comprising: a primary mast; a secondary mast slidably mounted to the primary mast; a garbage container holder slidably mounted to the secondary mast, a driving link connecting the primary mast to the secondary mast, and also connecting the garbage container holder to the secondary mast, a drive motor connected to drive the driving link in a manner that when the drive motor is operated, the driving link simultaneously slides the secondary mast along the primary mast, and slides the garbage container holder along the secondary mast, both sliding movements being in the same one of an upwards or downwards orientation, depending of a direction of operation of the drive motor.

[0007] In accordance with one aspect, there is provided a mechanical arm for a garbage collection truck performing automated garbage collection from a garbage bin. The mechanical arm comprises a grabber assembly, a vertical displacement section and a horizontal displacement section. The grabber assembly has a main body and wheels rotatably mounted to the main body and projecting therefrom. The grabber assembly is selectively engageable with the garbage bin to grasp the garbage bin, temporarily hold the garbage bin and subsequently release the garbage bin. The vertical displacement section comprises a multi-stage mast and a vertical actuation mechanism. The multi-stage mast extends substantially along a vertical axis and comprises a terminal mast section having a body with a terminal mast section guiding track defined therein. The terminal mast section guiding track includes a substantially straight lower portion and a consecutive curved upper portion. The grabber assembly is slidably movable on the terminal mast section with the wheels rotatably mounted to the main body of the grabber assembly being engaged in the terminal mast section guiding track. The vertical actuation mechanism drives sections of the multi-stage mast to slide onto one another and drives the grabber assembly to slide on the terminal mast section, with the wheels rotatably mounted to the main body of the grabber assembly moving along the guiding track of the terminal mast section. The horizontal displacement section extends substantially along a horizontal axis and is operatively connected to the grabber assembly to selectively move the grabber assembly substantially horizontally

[0008] In accordance with another aspect, there is provided a mechanical arm system for collecting garbage from a garbage container, the mechanical arm system comprising: a mast; a grabber assembly configured to selectively grab or release the garbage container; a sliding engagement between the grabber assembly and the mast, the sliding engagement providing a sliding path along which raising or lowering movement of the grabber assembly relative to the mast is guided, the sliding engagement including a first track provided along the mast, and a first wheel rotatably mounted to the grabber assembly, the first wheel engaged with the first track, a second track provided along the mast, the second track extending adjacent to the first track, and a second wheel rotatably mounted to the grabber assembly, the second wheel engaged with the second track, the first and second tracks having a lower portion being oriented in a vertical orientation, the grabber assembly being in front of the mast when the first wheel and second wheel are in the lower portion, the lower portion of the first and second tracks extending continuously to a respective upper portion of the first and second tracks, the upper portion of both the first and second tracks being curved rearwardly; an actuator to move the grabber assembly along the sliding path of the mast, in a manner that when the first wheel and second wheel reach and are moved along the upper portion of the corresponding tracks, the garbage container grabbed by the grabber assembly is tilted upside-down to empty its contents towards the rear.

[0009] In accordance with another aspect, there is provided a mechanical arm system for collecting garbage from a garbage container, the mechanical arm system comprising: a primary mast; a secondary mast; a first sliding engagement between the primary mast and the secondary mast, the first sliding engagement providing a first sliding path along which raising or lowering movement of the secondary mast along the primary mast is guided; a grabber assembly configured to selectively grab or release the garbage container; a second sliding engagement between the grabber assembly and the secondary mast, the second sliding engagement providing a second sliding path along which raising or lowering movement of the grabber assembly along the secondary mast is guided; linking chains connecting the grabber assembly to the secondary mast, and the secondary mast to the primary mast, a drive motor connected to the linking chains via a sprocket and operable in a manner to simultaneously raise or lower both i) the secondary mast along the primary mast via the linking chains and the first sliding engagement and ii) the grabber assembly along the secondary mast via the linking chains and the second sliding engagement.

[0010] Many further features and combinations thereof concerning the present improvements will appear to those skilled in the art following a reading of the instant disclosure.

Description

DESCRIPTION OF THE FIGURES

[0011] In the figures,

[0012] FIG. 1 is an oblique view of the mechanical arm shown secured to a garbage collection truck;

[0013] FIG. 2 is an exploded view of the mechanical arm of FIG. 1, shown without the corresponding garbage collection truck;

[0014] FIGS. 3A to 3G are perspective views of the mechanical arm, without the corresponding garbage collection truck, each one of the views showing the mechanical arm in a stage of a sequence of operation of the mechanical arm for collecting garbage;

[0015] FIG. 4 is another oblique view of the mechanical arm taken from the rear; and

[0016] FIG. 5 is a cross-sectional oblique view taken along a vertical and forward-rearward oriented cross-section plane from FIG. 4.

DETAILED DESCRIPTION

[0017] Referring generally to FIG. 1, in accordance with an embodiment, there is provided a mechanical arm **10** mountable to a garbage collection truck **14** to define an automated side loader (ASL) garbage truck performing automated collection of garbage from a garbage receptacle, generally referred to herein as a garbage bins **12** or garbage container. The mechanical arm **10** is designed to provide automated garbage collection with smooth, constant movement (i.e. movement that has minimal jerk/shock) thereby resulting in limited shock stress on the components and leading to high durability.

[0018] Referring to FIG. 2, in one embodiment, the mechanical arm **10** includes a grabber assembly **20** (alternately referred to herein as a garbage container holder) operative to engage and temporarily grasp successive corresponding garbage bins **12** during the garbage collection process. The grabber assembly **20** is a carriage operating as a prehension end-effector. The grabber assembly **20** is designed to grasp a garbage bin **12**, hold the garbage bin **12** as it is moved into the dumping position and brought back to its initial position, and then release the garbage bin **12** (i.e. operate generally as shown from FIGS. 3A to 3F and then back to 3A).

[0019] In the embodiment shown, the grabber assembly **20** is of the finger gripper type and includes a main body **22** with two opposed fingers **24a**, **24b** pivotally mounted to the main body **22** and extending therefrom. The fingers **24a**, **24b** each include a rotative finger joint **25**, thereby allowing the grabber assembly **20** to wrap around the garbage bin **12**, in order to securely grasp the

garbage bin **12** and adapt to several garbage bin sizes and shapes. The grabber assembly **20** is configurable between an open configuration where the fingers **24a**, **24b** extend opposed to one another and define a bin receiving space **26** in which a garbage bin **12** can be positioned to be subsequently grasped (see FIG. **3A**) and a closed configuration where the fingers **24a**, **24b** are brought towards one another to engage the garbage bin **12** and hold it in a secure grip (see FIG. **3C**). In the embodiment shown, in the closed configuration, the fingers **24a**, **24b** are wrapped over the garbage bin **12**, but one skilled in the art will understand that, in an alternative embodiment (not shown) the fingers **24a**, **24b** could simply engage the sides of the garbage bin and press thereagainst to ensure a secure grip.

[0020] In this embodiment, the grabber assembly **20** also includes grabber actuators **23** connected between the main body **22** and the sections of the fingers **24a**, **24b** and operative to move the fingers **24a**, **24b** between the open configuration (see FIGS. **3A**) and the closed configuration (see FIG. **3B**). In the embodiment shown, the grabber actuators **23** include a combination of rotary actuators and linear actuators, such as pneumatic cylinders, hydraulic cylinders, electro-mechanical actuators, or the like. One skilled in the art will understand that, in alternative embodiments, actuators different from the combination of rotary and linear actuators could also be used. In view of the above, one skilled in the art will understand that the grabber actuators **23** can be embodied using either one of electrical actuators or hydraulic actuators. Moreover, in an embodiment where hydraulic actuators are initially used for the grabber actuators **23**, the actuators can subsequently be replaced to switch to electric actuators and vice-versa.

[0021] In the embodiment shown, end sections **26a**, **26b** of the fingers **24a**, **24b** have a different configuration between the first finger **24a** and the second finger **24b**. The first finger **24a** has two members **27a**, **27b** spaced apart from one another and defining an opening **28** therebetween. The second finger **24b** has a single member **29** positioned to extend in the opening between the two members **27a**, **27b** of the first finger **24a** when the fingers **24a**, **24b** are configured in the closed configuration. In the embodiment shown, the two members **27a**, **27b** of the end section **26a** of the first finger **24a** are also of a longer length than the single member **29** of the end section **26b** of the second finger **24b**. In the embodiment shown, the fingers **24a**, **24b** have an inner surface, with the inner surface being lined with resilient anti-slip material, such as rubber pads **21** or the like.

[0022] One skilled in the art will understand that, in an alternative embodiment (not shown), the grabber assembly **20** could present a different configuration and/or a different operating mechanism than the grabber assembly **20** of the embodiment shown, while still allowing the required grasp, temporary hold and subsequent release of the garbage bin **12**.

[0023] The mechanical arm **10** also includes a vertical displacement section **30**. The grabber assembly **20** is operatively mounted to the vertical displacement section **30** and cooperates therewith to move the grabber assembly **20** along a vertical axis Y and selectively tilt (or tumble) the grabber assembly **20** with regard to the vertical axis Y. The tilting of the grabber assembly **20** with regard to the vertical axis Y allows the dumping of the content of the garbage bin **12** grasped and temporarily maintained by the grabber assembly **20**, into the collection container **16** of the garbage truck **14** (or a section thereof).

[0024] The vertical displacement section **30** includes a multi-stage mast **32** extending substantially along the vertical axis Y. In the embodiment shown, the multi-stage mast **32** is a two stage (or duplex) mast **32** having a first mast section **40** (also referred to herein as primary mast) and a second mast section **50** (also referred to herein as terminal mast section or secondary mast). The second mast section **50** is slidably mounted to the first mast section **40** and is linearly and slidably movable thereabout. The main body **22** of the grabber assembly **20** is slidably mounted to the second mast section **50** and is slidably movable therealong to perform a portion of the vertical displacement and the tilting of the grabber assembly **20**.

[0025] One skilled in the art will understand that, in an embodiment, the vertical axis Y can be perfectly vertical. In an alternative embodiment, the vertical axis Y can be angled with respect to a

perfectly vertical axis. Therefore, the multi-stage mast **30** of the vertical displacement section **30** could extend perfectly vertical or substantially vertical (i.e. generally vertically, without having a perfectly vertical orientation), in a context of a truck standing on a perfectly horizontal surface. The orientation can then change when the truck is positioned on sloping ground, for instance.

[0026] As best seen in FIG. 2, the first mast section **40** includes an elongated body **42** extending substantially along the vertical axis Y, with a guiding track **44** (or first mast section guiding tracks) defined therein. In the embodiment, the guiding track **44** consists of two guiding tracks **44** positioned on opposite lateral faces of the body **42**, which extend generally vertically along the vertical axis Y. In the embodiment shown, the guiding tracks **44** of the first mast section **40** have a slightly concave curvilinear profile. In other words, in the embodiment shown, the guiding tracks **44** of the first mast section **40** define a slight curve toward the exterior of the vehicle, generally arcuate in shape. One skilled in the art will understand that, in an alternative embodiment (not shown), the guiding tracks **44** of the first mast section **40** could have a substantially rectilinear profile, rather than the concave curvilinear profile as shown.

[0027] The second mast section **50** also includes an elongated body **52** extending substantially along the vertical axis Y. The elongated body **52** of the second mast section **50** includes opposed lateral walls **54a**, **54b** spaced apart from one another and configured to receive the body **42** of the first section **40** of the mast **32** therebetween. Hence, the vertical side walls **54a**, **54b** of the elongated body **52** of the second mast section **50** extend on opposed sides of the first mast section **40**. The second mast section **50** also includes wheels (not shown) rotatably mounted to an inner surface of the vertical side walls **54a**, **54b**, each wheel being engaged in a corresponding guiding track **44** of the first mast section **40**. The wheels can roll within the guiding tracks **44** to allow the second mast section **50** to slide smoothly, with limited friction, vertically with regard to the first mast section **40**. In an embodiment where the secondary mast is symmetrical along a medial plane (vertical and forward-rearward oriented), the adjacent tracks on a first side of the secondary mast can be referred to as the first set of tracks, and the adjacent tracks on a second side of the secondary mast can be referred to as the second set of tracks, whereas the wheels on a first side of the garbage container holder can be referred to as a first set of wheels engaged with the first set of tracks, and the wheels on a second side of the garbage container holder can be referred to as a second set of wheels engaged with the second set of tracks.

[0028] In the embodiment shown, the concave curvilinear profile of the guiding tracks **44** of the first mast section **40**, results in the second mast section **50** being lightly tilted rearwardly, as the second mast section **50** is moved upwardly with regard to the first mast section **40**, as shown in FIG. 3E. One skilled in the art will understand that, in an alternative embodiment (not shown), where the guiding tracks **44** of the first mast section **40** would have a substantially straight profile, the second mast section **50** could remain in the same orientation as it is moved up and down along the first mast section **40**.

[0029] The second mast section **50** includes two generally parallel guiding tracks **56** which will be referred to as an inner track **57** and an outer track **58**. In this embodiment, both the inner track **57** and the outer track **58** include two laterally opposite track segments, one on each lateral side of the second mast section. In this embodiment, both guiding tracks **57**, **58** are roughly parallel, and can be said to have a lower portion and an upper portion. The lower portion **56a** is straight and extends substantially linearly along the vertical axis Y. The upper portion **56b** is curved. The lower portion **56a** and the upper portion **56b** are consecutive and extend continuously, forming a continuous wheel path. The upper portion **56** is positioned at an upper end of the corresponding one of the opposed vertical side walls **54a**, **54b** and has a pronounced curved profile, with the curve leading inwardly, i.e. in a direction towards a rear of the arm **10**.

[0030] In an embodiment, being adjacent one another, the degree of curvature of the curved upper portion **56b** of the inner track **57** is greater than the degree of curvature of the corresponding outer track **58**. In other words, the radius of the upper portion **56b** of the inner track **57** is smaller than the

curve radius of the upper portion of the outer track **58**. The grabber assembly **20** has two sets of wheels, an upper set engaged with the inner track **57**, and a lower set engaged with the outer track **58**. When the wheels of the grabber assembly **20** are slid along the corresponding tracks **57**, **58**, eventually along the curved upper portions thereof and to a limit of the wheel path, the upper wheels are pivoted around the smaller radius of the inner track **57**, whereas the lower wheels are pivoted around the greater radius of the outer track **58**, which leads to a significant (but relatively smooth) change in orientation of the grabber assembly **20**, to the extent where its load is tilted rearwardly and upside-down, directing the contents of the bin into the truck's chamber.

[0031] In view of the above, in an embodiment, the curve radius of the curved upper portion **56b** of the inner track segment **57** is configured to provide a smooth and relatively continuous pivoting path offering limited jerk of the grabber assembly **20** during tilting. The curve radius of the curved upper portion **56b** of the inner track segment **57** can also, as shown, be configured to avoid interference between the grabber assembly **20** and the second mast section **50**. It will be understood that a curve radius of the upper portion **56b** of the inner track **57** that is too great could result in the grabber assembly **20** tilting too early and colliding with the second mast section **50**, thereby preventing proper operation of the arm **10**. In other words, the combination of the position of the upper portion **56b** of the inner track **57** with regard to the substantially straight lower portion **56a** of the corresponding guiding track **56** and the curve radius of the upper portion **56b** of the inner track **57**, is such that no interference occurs between the grabber assembly **20** and the second mast section **50**, as the grabber assembly **20** tilts.

[0032] In the embodiment shown, the curve radius of the outer track **58** is set in accordance with the position of pinions **36** of a vertical actuation mechanism **34** (which will be described in more details below), such that when the grabber assembly **20** is driven in the curved upper portion **56b** of the guiding tracks **56**, a linkage **38** connecting the grabber assembly **20** to heavy-duty roller chains **35** engaged with the pinions **36** of the vertical actuation mechanism **34** remains substantially in line with a tangent of the section of the chains **35** engaged with the corresponding pinions **36**, thereby also contributing to minimizing acceleration of the grabber assembly **20** during tilting.

[0033] The main body **22** of the grabber assembly **20** also includes vertical side walls **19a**, **19b** spaced apart from one another and configured to receive the body **52** of the second mast section **50** therebetween. The vertical side walls **19a**, **19b** of the main body **22** of the grabber assembly **20** extend on opposed sides of the second mast section **50**. The grabber assembly **20** also includes wheels **18** rotatably mounted to an inner surface of the vertical side walls **19a** **19b**, and engageable in the corresponding guiding track **57**, **58** of the second mast section **50** to allow the grabber assembly **20** to move vertically with regard to the second mast section **50**. In the embodiment shown, two wheels **18** are rotatably mounted to the inner surface of each one of the vertical side walls **19a** **19b**, one of these two wheels being engageable in a respective one of the inner track **57** and the outer track **58**.

[0034] In view of the above, the grabber assembly **20** initially holds the bin **12** in a natural upright orientation and moves it upwardly along the second mast section **50** (with the wheels **18** rolling in the lower portions of the corresponding tracks and subsequently tilts the bin rearwardly when it reaches the upper end of the second mast section **50**, i.e. when the wheels **18** travel into the curved upper portion **56b** of the corresponding guiding track **56** of the second mast section **50**.

[0035] In the embodiment shown, the wheels **18** engaged in the inner track **57** are positioned upwardly from the wheels **18** engaged in the outer track **58** on the vertical side walls **19a** **19b** of the grabber assembly **20**. When combined with the inner track **57** having a greater degree of curvature in the upper portion **56b** than the corresponding portion of the outer track **58**, such a positioning of the wheels **18** result in a smoother tilting of the grabber assembly **20** (and the garbage bin **12** maintained thereon). Indeed, in the embodiment shown, when the grabber assembly is raised along the tracks **57**, **58**, the wheels **18** engaged in the inner track **57** reach the curved upper portion of the inner track **57** and begin to pivot before the wheels **18** engaged in the outer track **58** has reached the

corresponding curved upper portion. This smooth, continuous motion can lead to a smooth tilting of the grabber assembly **20** and efficient dumping of the content of the garbage bin **12** grasped by the grabber assembly **20**, while minimizing potential mechanical fatigue failure of the components. [0036] The vertical displacement section **30** also includes a vertical actuation mechanism **34** operative to vertically displace the second mast section **50** and the grabber assembly **20**. In other words, the vertical actuation mechanism **34** is operative to drive the mast sections of the multi-stage mast **32** to slide onto one another and to drive the grabber assembly **20** to slide on the second mast section **50** (or terminal mast section).

[0037] In the embodiment shown, the actuation mechanism **34** includes heavy-duty roller chains **35** having a portion securable to the first mast section **40** and extending between the above-referenced pinions **36** rotatably mounted to the body **52** of the second mast section **50**. The heavy-duty roller chains **35** are actuatable by a rotary actuator **37** selectively driving at least one of the pinions **36** in rotation. In the embodiment shown, the rotary actuator **37** is connected to a pinion **36** positioned at the lower end of the body **52** of the second mast section **50**, but one skilled in the art will understand that, in alternative embodiments (not shown), the rotary actuator **37** could be connected to any of the pinions **36**. The grabber assembly **20** is also securable to a portion of the heavy-duty roller chains **35**, through the above referenced linkage **38** securely connecting the grabber assembly **20** to the heavy-duty roller chains **35**, in order to move therewith.

[0038] One skilled in the art will understand that the rotary actuator **37** can be embodied using either one of an electrical actuator or an hydraulic actuator. Moreover, in an embodiment where an hydraulic actuator is initially used for the rotary actuator **37**, the actuator can subsequently be replaced to switch to an electric actuator and vice-versa. In this embodiment, electric actuators were preferred.

[0039] In the embodiment shown, the vertical actuation mechanism **34** includes two heavy-duty roller chains **35**, but one skilled in the art will understand that, in an alternative embodiment (not shown), a single roller chain or more than two roller chains could be used. In an embodiment, the first mast section **40** includes projections **46** each securable to the portion of the corresponding heavy-duty roller chains **35**, for example by engaging one of the links of the corresponding roller chain **35** therewith.

[0040] In operation, given that a portion of the heavy-duty roller chains **35** is secured to the first mast section **40**, the actuation of one of the pinions **36** by the rotary actuator **37** results in the second mast section **50** being moved upwardly/downwardly with respect to the first mast section **40** (i.e. the second mast section sliding upwardly/downwardly with respect to the first mast section **40**). In view of the above, the upward/downward movement of the second mast section **50** with respect to the first mast section **40**, drives the wheels (not shown) of the second mast section **50** along the guiding tracks **44** of the first mast section **40**.

[0041] Given that the grabber assembly **20** is also secured to a portion of the heavy-duty roller chains **35**, the actuation of one of the pinions **36** by the rotary actuator **37** simultaneously results in the grabber assembly **20** being moved upwardly/downwardly with respect to the second mast section **50** (i.e. the grabber assembly **20** sliding upwardly/downwardly on the second mast section **50**). As mentioned above, the upward movement of the grabber assembly **20** with respect to the second mast section **50**, drives the wheels **18** of the grabber assembly **20** along the guiding tracks **56** of the second mast section **50** and moves the grabber assembly **20** substantially straight as the wheels **18** are engaged in the substantially straight lower portion **56a** of the corresponding guiding track **56** and in a rearward tilting movement as the wheels **18** travel into the curved upper portion **56b** of the corresponding guiding track **56**. The reverse movements are performed during the downward movement of the grabber assembly **20** with respect to the second mast section **50**.

[0042] In an embodiment, the vertical actuation mechanism **34** also includes a chain tensioning mechanism (not shown) operative to adjust the tension of the roller chains **35**. For example and without being limitative, in an embodiment, the chain tensioning mechanism (not shown) can

include at least one linear actuator, such as a pneumatic cylinder, an hydraulic cylinder, an electro-mechanical actuator, or the like, operatively connected to a corresponding one of the pinions **36** to move the pinion **36** and consequently adjust the tension of the corresponding roller chain **35**. In an embodiment, the chain tensioning mechanism (not shown) is operatively connected to the pinion **36** actuated by the rotary actuator **37**.

[0043] One skilled in the art will understand that, in alternative embodiments, a vertical actuation mechanism different from the vertical actuation mechanism **34** of the embodiment shown could also be used to vertically displace the second mast section **50** and the grabber assembly **20** (i.e. move the second mast section **50** and the grabber assembly **20** in up/down movements). For example and without being limitative, linear actuators such as pneumatic cylinders, hydraulic cylinders, electro-mechanical actuators, or the like, or a combination of linear actuators and roller chains with corresponding pinions could be used to operatively connect the first mast section **40** and the second mast section **50** and the second mast section **50** and the grabber assembly **20**. In view of the above, one skilled in the art will understand that the either one of electrical actuators or hydraulic actuators could be used and that, in an embodiment where hydraulic actuators are initially used for, the actuators can subsequently be replaced to switch to electric actuators and vice-versa. Other driving mechanism could also be used.

[0044] In an embodiment, the second mast section **50** includes a support hook (not shown) extending therefrom, at the upper end thereof, and the grabber assembly **20** includes a projection (not shown) engageable with the support hook (not shown) of the second mast section **50**, when the grabber assembly **20** reaches the upper end of the guiding tracks **56** of the second mast section **50**. The combination of the hook (not shown) of the second mast section **50** and the projection (not shown) of the grabber assembly **20**, allows reduction of the strain on the chains **35**, when the grabber assembly **20** has reached the upper most position and tilts to dump the garbage, a portion of the weight of the grabber assembly **20** and the garbage bin **12** being supported by the hook (not shown) engaged in the projection (not shown). In other words, the combination of the hook (not shown) of the second mast section **50** and the projection (not shown) of the grabber assembly **20** supports a portion of the weight of the grabber assembly **20** and the garbage bin **12**, during the dumping of the garbage from the garbage bin **12**, when the hook (not shown) is engaged in the corresponding projection (not shown). One skilled in the art will understand that, in an alternative embodiment, the hook and projection could be inverted, i.e. the support hook (not shown) could extend from the main body **22** of the grabber assembly **20**, with the projection projecting from the body **52** of the second mast section **50**. One skilled in the art will understand that, in an alternative embodiment, the mechanical arm **10** could be free of the above described hook and projection assembly.

[0045] One skilled in the art will also understand that, in an alternative embodiment (not shown), the vertical displacement section **30** could be different from the vertical displacement section **30** of the embodiment shown. For example and without being limitative, the multi-stage mast **32** could include more than two stages (i.e. the vertical displacement section **30** could include a triple stage mast, a quadruple stage mast, etc.), with the last mast section (i.e. the mast section moving the highest when the mast **32** is completely deployed) being the terminal mast section.

[0046] In an embodiment, the mechanical arm **10** further includes a horizontal displacement section **60** extending substantially along an horizontal axis X and being operatively connected to the grabber assembly **20** to selectively move the grabber assembly **20** substantially horizontally. In other words, the horizontal displacement section **60** is operative to move the vertical displacement section **30** of the mechanical arm **10**, substantially along a horizontal axis X, thereby producing in and out movements of the grabber assembly **20** with regards to the garbage collection truck **14** and allowing the garbage bins **12** located within a horizontal range of the garbage collection truck **14** to be engaged by the grabber assembly **20**.

[0047] In the embodiment shown, the horizontal displacement section **60** includes a connecting

shaft **62** with the first mast section **40** (i.e. the multi-stage mast **32**) of the vertical displacement section **30** secured thereto, at a distal end thereof. The connecting shaft **62** is horizontally movable along the horizontal axis X, to move the vertical displacement section **30** (and the grabber assembly **20** positioned at the front end thereof) horizontally in in/out movements with regard to the garbage collection truck **14**. The connecting shaft **62** is slidably mounted to a longitudinal horizontal rail **64** and is longitudinally displaceable thereabout. The horizontal rail **64** is secured to the garbage collection truck **14** and includes a sliding mechanism **65** with sliding components such as tracks and free rolling rollers to minimize friction between the horizontal rail **64** and the longitudinally displaceable connecting shaft **62**.

[0048] One skilled in the art will understand that, in an embodiment, the horizontal axis X can be perfectly horizontal. In an alternative embodiment, the horizontal axis X can be angled upwardly or downwardly with respect to a perfectly horizontal axis. Therefore, the horizontal displacement section **60** could perform perfectly horizontal movement, or substantially horizontal movement, which vary from a perfectly horizontal orientation, while remaining generally horizontal.

[0049] In an embodiment, the horizontal displacement section **60** further includes a horizontal actuation mechanism **66** operatively connected between the connecting shaft **62** and the horizontal rail **64**, to move the connecting shaft **62** about the horizontal rail **64**. In the embodiment shown, the actuation mechanism **66** includes a rotary actuator **67** driving a pinion (not shown) of a rack and pinion assembly **68**. One skilled in the art will understand that, in an alternative embodiment (not shown), the horizontal actuation mechanism **66** could be different from the embodiment shown. For example and without being limitative, in an embodiment (not shown), the horizontal actuation mechanism **66** could include a linear actuator. Once again, one skilled in the art will understand that the rotary actuator **67** or alternative linear actuator can be embodied using either one of an electrical actuator or an hydraulic actuator and that, in an embodiment where an hydraulic actuator is initially used, the actuator can subsequently be replaced to switch to an electric actuator and vice-versa.

[0050] One skilled in the art will understand that, in an alternative embodiment (not shown), the horizontal displacement section **60** could be different from the horizontal displacement section **60** of the embodiment shown while still allowing the vertical displacement section **30** of the mechanical arm **10** to be moved substantially along the horizontal axis X and produce the desired in/out movements of the grabber assembly **20** with regards to the garbage collection truck **14**.

[0051] In an embodiment, the mechanical arm **10** can include a weight sensor (not shown), such as, without being limitative, a strain gage weight sensor. In an embodiment, the weight sensor (not shown) is mounted between the second mast section **50** and the grabber assembly **20**, thereby allowing the measure of the weight of individual garbage bins **12**, as they are being grasped by the mechanical arm **10**. The minimal swing-out and constant movement (i.e. movement that has minimal jerk) provided by the mechanical arm **10** described herein, can help in acquiring accurate weight data for the weight of the individual garbage bins **12** using the weight sensor (not shown).

[0052] In an embodiment, the mechanical arm **10** includes an electronic control system (not shown) operatively connected to the actuators **23**, **37**, **67** of the grabber assembly **20** and/or the vertical displacement section **30** and/or the horizontal displacement section **60**, such that the operations of the components thereof for grasping, moving and releasing the garbage bin **12** are synchronized. In an embodiment, the electronic control system (not shown) can move the components with a different velocity, depending of the position. For example and without being limitative, the velocity of the components of the vertical displacement section **30** can be greater when the second mast section **50** and the grabber assembly **20** are driven in a substantially straight direction than when the grabber assembly **20** is being tilted (i.e. when the wheels **16** of the grabber assembly **20** are moving into the curved portion **56b** of the guiding tracks **56**).

[0053] Referring to FIGS. 3A to 3F, the mechanical arm **10** therefore allows the automated collection of garbage located in garbage bins **12** left for collection by users. In order to perform

garbage collection, the mechanical arm **10** is initially extended outwardly (i.e. moved outwardly from the garbage collection truck **14** in a substantially horizontal movement, between a retracted configuration and an outwardly extended configuration) (see FIGS. **3A** and **3B**, with the mechanical arm **10** shown in a retracted configuration in FIG. **3A** and in an outwardly extended configuration in FIG. **3B**). The mechanical arm **10** is extended outwardly through operation of the horizontal displacement section **60**.

[0054] In the outwardly extended configuration, the grabber assembly **20** engages a collection bin **12**. The grabber assembly **20**, can subsequently be moved from the open configuration to the closed configuration, to grasp and hold the collection bin **12** (see FIGS. **3B** and **3C**, with the grabber assembly **20** shown in the open configuration in FIG. **3B** and in the closed configuration in FIG. **3C**) and the mechanical arm **10** can be retracted inwardly (i.e. moved inwardly towards the garbage collection truck **14** in a substantially horizontal movement, between the outwardly extended configuration and the retracted configuration) (see FIGS. **3C** and **3D**, with the mechanical arm **10** shown grasping the garbage bin **12** in the outwardly extended configuration in FIG. **3C** and in the retracted configuration in FIG. **3D**).

[0055] Subsequently, the vertical displacement section **30** is used to move the second mast section **50** and the grabber assembly **20** upwardly, with the grabber assembly **20** securely holding the garbage bin **12** (see FIGS. **3E** and **3F**, wherein the vertical displacement section **30** is shown in an intermediary raised configuration in FIG. **3E** and the vertical displacement section **30** is shown in a completely raised configuration in FIG. **3F**. As can be seen in FIG. **3F**, when the vertical displacement section **30** is brought to the completely raised configuration, the grabber assembly **20** is tilted to empty the content of the garbage bin **12** hold by the grabber assembly **20**. The tilting of the grabber assembly **20** results from the above described grabber assembly **20** having wheels **18** engaged in the guiding tracks **56** of the second mast section **50** and thereby following the path of the guiding tracks **56** as the grabber assembly is slid onto the second mast section **50**, with the track having a substantially straight lower portion **56a** and a curved upper portion **56b**. As mentioned above, the curved upper portion **56b** of the guiding tracks **56** has a degree of curvature sufficient to cause the tilting (or tumbling) of the garbage bin **12** hold by the grabber assembly **20** and resulting in emptying of the content thereof.

[0056] The reverse operation sequence can subsequently be performed to move the second mast section **50** and the grabber assembly **20** downwardly (see FIGS. **3F** and **3E**), drive the grabber assembly **20** and the corresponding garbage bin **12** outwardly (see FIG. **3D**), release the garbage bin **12** (see FIG. **3C**) and bring back the mechanical arm **10** in the retracted configuration (See FIGS. **3B** and **3A**).

[0057] Even though the term “garbage” is used herein, it will be understood that the mechanical arm **10** can be used to perform collection of any type of material which can be placed in a bin **12** (or container) that can be manipulated by the mechanical arm **10**, such as, without being limitative, municipal solid waste (or trash), recycling material, food waste (or organic waste), or the like

[0058] As can be understood, the examples described above and illustrated are intended to be exemplary only. The scope is indicated by the appended claims.

Claims

1. A mechanical arm system for collecting garbage from a garbage container, the mechanical arm system comprising: a garbage container holder having two guiding wheels at a rear thereof, the garbage container being held at a front of the garbage container holder; a mast having two adjacent tracks each receiving a corresponding wheel to guide the movement thereof, the two adjacent tracks both having a lower portion extending upwardly along the mast, the lower portion of the first track being located forwardly of the lower portion of the second track, the lower portion of both tracks leading into corresponding upper portions which are curved rearwardly in a manner that the

garbage container holder can be raised along the lower portion of the tracks and into the upper portion of the tracks, the garbage container being tilted upside down to empty its contents behind the mast as it is conveyed along the upper portions of the tracks.

2.-19. (canceled)
